

A PROJECT REPORT
ON
**“An AI-Based Simulation System for Reconstructing
Ancient Indian Bead Numeration Algorithms”**

IT Project Management
(21MGH303P)

Submitted to the partial fulfilment of the
requirement for the award of the degree

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

This project, titled “An AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms,” presents the design and development of an AI-powered educational web application aimed at preserving and explaining the historical foundations of ancient Indian numeration and their relationship to modern digital computing. The system was created to address the lack of interactive educational platforms that combine historical mathematical knowledge, simulation-based learning, and artificial intelligence in a unified environment.

The proposed platform integrates AI-driven narration, an interactive abacus-style bead simulator, ancient calculation methods, a number system explorer, and a module explaining the connection between zero and binary logic. The application is designed to help learners understand the evolution of Indian mathematics through engaging visual and conceptual learning experiences rather than through static textual explanation alone.

Technically, the system is implemented as a web-based platform using Next.js and React for the frontend, Firebase for authentication and data management, Gemini for AI-based educational narration, and Vercel for deployment. The architecture follows a layered client-server design that supports scalability, maintainability, secure access, and smooth integration between educational content, simulation logic, and AI services.

The project was developed through structured software project management practices covering requirement analysis, feasibility assessment, scheduling, budgeting, testing, risk management, and project closure. Functional modules such as AI narration, abacus simulation, quiz assessment, progress tracking, and responsive user interaction were developed and validated through test cases and quality metrics.

The final outcome demonstrates that AI can be effectively used to reconstruct and communicate traditional mathematical concepts in an accessible and modern form. The project contributes to digital heritage preservation, educational technology, and interdisciplinary learning by showing how ancient Indian bead numeration algorithms and the concept of zero can be meaningfully connected to present-day computational thinking.

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INTRODUCTION

The project titled “An AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms” is an educational technology initiative designed to preserve and present the rich mathematical heritage of ancient India through an interactive digital platform. The system aims to bridge the gap between historical knowledge and modern computational learning by combining simulation, storytelling, and artificial intelligence in a single web-based environment.

The main purpose of this project is to help learners understand ancient Indian numeration methods, the significance of zero, and the evolution of number systems such as decimal, octal, hexadecimal, and binary. Instead of relying only on conventional textbook explanations, the platform provides a more engaging experience through interactive modules, visual demonstrations, and AI-generated narration that simplifies complex historical and mathematical concepts.

The key highlights of the project are as follows:

- AI-powered narration for explaining the history and significance of ancient Indian numeration.
- Interactive abacus-style bead simulator for demonstrating arithmetic operations and bead-based algorithms.
- Modules for ancient calculation techniques, number system conversion, and the connection between zero and binary logic.
- Web-based implementation using Next.js and React, with Firebase for backend services and Gemini for intelligent content generation.
- A responsive and educational platform intended for students, educators, and learners interested in mathematics, history, and computing.

Overall, the project demonstrates how modern AI tools can be used to reconstruct, preserve, and teach traditional mathematical ideas in a meaningful and accessible way. It also highlights the educational and cultural importance of connecting ancient Indian knowledge systems with present-day digital learning technologies.

UNIT 1

Introduction to Project Management

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

1.1 Experiment 1: Project Requirement Gathering and Analysis

1.1.1 Software Project Management and Scope

Software project management encompasses the systematic planning, organizing, directing, monitoring, and controlling of software development projects to ensure they are completed on time, within budget, and to the required quality standards. It bridges the gap between technical development and business objectives by applying structured methodologies to manage people, processes, and technology effectively.

The AI-Based Bead Numeration Simulation System project scope encompasses the full-cycle development of an AI-powered educational web application that teaches users about the ancient Indian numeration system, the mathematical significance of zero, the evolution of number systems (decimal, octal, hexadecimal, binary), and the foundational role of Indian mathematics in modern digital computing. The system leverages Next.js for the frontend, Firebase for backend services, Gemini LLM for AI-driven story narration, and is deployed on Vercel. The project is managed under a structured five-phase plan spanning January 2026 to May 2026, executed by a three-member team.

1.1.2 Requirement Engineering and Elicitation

Requirement engineering is the disciplined process of eliciting, analyzing, documenting, prioritizing, and validating the needs and constraints of all stakeholders before and during software development. It ensures that the final system addresses real user needs and aligns with the project's academic and educational objectives.

For this project, requirement gathering was conducted using a hybrid approach combining facilitated workshops, document analysis, and competitive analysis. The team members included Ashray Singh (Project Manager), Madhav Nehra (Full-Stack Developer), and Rudransh Yadav (AI and Content Engineer), along with the faculty advisor for academic validation.

1.1.3 Functional Requirements

ID	Requirement Description	Source	Priority
FR01	AI-powered story narration of Indian numeration history using Gemini LLM	Workshop	High
FR02	Interactive abacus simulator with bead manipulation for addition and subtraction	Workshop	High
FR03	Ancient Indian calculation methods module (Vedic Math tricks, historical methods)	Document Analysis	High
FR04	Number system explorer with conversion tool (Decimal ↔ Octal ↔ Hex ↔ Binary)	Workshop	High
FR05	Zero and Binary connection module explaining Boolean logic and digital circuits	Document Analysis	High
FR06	Quiz and assessment module for student self-evaluation	Workshop	Medium
FR07	User authentication and role-based access using Firebase Auth	Technical Review	Medium
FR08	Admin and content management dashboard for educators	Workshop	Medium
FR09	Progress tracking for registered students across modules	Technical Review	Medium
FR10	Responsive UI accessible on desktop and mobile browsers	Document Analysis	High

1.1.4 Non-Functional Requirements

Category	Requirement	Target Metric	Priority
Performance	Page load time for all modules	< 3 seconds	High
AI Response Time	Gemini narration generation latency	< 5 seconds	High
Scalability	Concurrent users supported	Up to 500	Medium
Availability	Platform uptime	99% uptime	Medium
Security	Firebase Auth data protection	Role-based access	High
Maintainability	Modular codebase for updates	New module in 3 person-days	Medium
Usability	Learning curve for new student users	< 30 minutes	High
Compatibility	Browser and device compatibility	Chrome, Firefox, Edge	High
Reliability	Mean Time Between Failures	> 150 hours	Medium

1.1.5 Requirement Prioritization using MoSCoW Method

The MoSCoW prioritization framework was applied to classify all requirements into Must Have, Should Have, Could Have, and Won't Have categories based on delivery criticality and timeline constraints.

Must Have requirements included AI story narration, interactive abacus simulator, number system explorer, zero-binary module, user authentication, and responsive UI. Should Have requirements included Vedic Math methods, quiz module, and progress tracking. Could Have features included multilingual support, gamification, and offline mode. Mobile-native apps and ERP integrations were considered out of scope.

1.1.6 Requirements Traceability Matrix

Req ID	Source	Business Driver	Validation Method
FR01	Workshop	AI-driven educational engagement	Narration test with students
FR02	Workshop	Hands-on ancient tool simulation	20 bead operation scenarios
FR03	Document Analysis	Preserving historical computation methods	Faculty expert review
FR04	Workshop	Bridge ancient and modern number systems	50 conversion test cases
FR05	Document Analysis	Link Indian mathematics to digital architecture	Domain review
FR06	Workshop	Student self-assessment	UAT with student users
FR07	Technical Review	Secure multi-role access	Firestore Auth integration test
FR08	Workshop	Content management for educators	Admin panel walkthrough
FR09	Technical Review	Track student learning progress	Firestore test with records
FR10	Document Analysis	Broad device accessibility	Cross-browser testing

1.1.7 Project Scope Management

In scope are the AI narration module, interactive abacus simulator, Vedic mathematics content, number system explorer, zero and binary module, quiz and self-assessment, Firebase authentication, Vercel deployment, and complete user documentation. Out of scope are native mobile app development, live tutoring, LMS integration, OCR features, and curriculum beyond the defined modules.

1.2 Experiment 2: Project Identification Process Methodology and Stakeholder Description

1.2.1 Project Identification and Selection

Phase	Activity	Outcome for This Project
Problem Identification	Analysis of gaps in educational tools	Absence of AI-interactive Indian numeration tools identified
Opportunity Analysis	Assessment of AI storytelling potential	Gemini suitable for historical narration
Feasibility Study	Technical, economic, and operational assessment	Project found feasible
Project Justification	Alignment with educational objectives	Supports IT Project Management curriculum
Approval Gate	Faculty submission and approval	Approved by faculty advisor

1.2.2 Feasibility Study Details

Technical feasibility was confirmed through the compatibility of Next.js, Firebase, Gemini API, and Vercel deployment. Economic feasibility estimated the total project cost between INR 8–12 Lakhs, while operational feasibility was supported by the availability of a skilled three-member team and a faculty advisor.

1.2.3 Initiation Phases and Contracting

The project initiation involved Project Charter development, stakeholder identification, role assignment, technology stack agreement, and timeline commitment. Ashray Singh was designated as Project Manager, Madhav Nehra as Full-Stack Developer, and Rudransh Yadav as AI and Content Engineer.

1.2.4 Stakeholder Analysis

Stakeholder	Role in Project	Interest Level	Power / Influence	Expectation
Faculty Advisor	Academic oversight and grading	High	High	On-time delivery and proper documentation
Ashray Singh	Planning, scheduling, reporting	High	High	Successful project delivery
Madhav Nehra	Frontend and backend development	High	Medium	Clear requirements and architecture
Rudransh Yadav	Gemini integration and educational content	High	Medium	Well-defined AI scope
Students	Primary learners using platform	High	Low	Engaging and accurate content
Educators	Secondary users managing content	Medium	Low	Easy content management
College/Institution	Academic value and compliance	Medium	High	Demonstrated learning outcomes

1.2.5 Power-Interest Grid Classification

Quadrant	Stakeholders	Engagement Strategy
High Power, High Interest	Faculty Advisor, College	Manage Closely
High Power, Low Interest	College Administration	Keep Satisfied
Low Power, High Interest	Students, Educators	Keep Informed
Low Power, Low Interest	General Public	Monitor

1.2.6 Stakeholder Engagement Strategy

Stakeholder	Engagement Method	Frequency
Faculty Advisor	Milestone review + report submission	Bi-weekly
Ashray Singh	Internal team stand-up	Daily
Madhav Nehra	Sprint planning + code review	Per Sprint
Rudransh Yadav	Content and AI validation sessions	Weekly
Students	Beta testing + feedback surveys	Per module release
Educators	Demo walkthrough + usability feedback	At v1.0 release

1.3 Experiment 3: Project Cost Estimation and Capital Budgeting

1.3.1 Cost Components Breakdown

Cost Category	Description	Estimated Range (INR)
Development	Frontend, backend, AI integration, module development	4–6 Lakhs
AI & Content	Gemini usage and content design	1–1.5 Lakhs
Infrastructure	Vercel hosting, Firebase plan, domain	0.5–1 Lakh
Tools & Licenses	GitHub Pro, Figma, tools	0.25–0.5 Lakhs
Testing & QA	Functional and cross-browser testing	0.5–1 Lakh
Project Management	Documentation and reporting	0.5 Lakh
Contingency (15%)	Unforeseen expenses	1–1.5 Lakhs
Total		8–12 Lakhs

1.3.2 Software Cost Estimation Models

COCOMO Intermediate Organic Mode was used because this is a relatively small project with a three-member team operating in a familiar technology environment. Using a midpoint estimate of 20 KLOC, the project effort was estimated at approximately 78.6 person-months before adjustment. A practical adjusted effort range of 40–50 person-months was adopted for the academic setting with parallel task execution.

Function Point Analysis produced an adjusted value of approximately 127 FP, which was further interpreted as 150–180 FP considering AI complexity and simulation-heavy modules. Expert judgment from the team validated a budget estimate of INR 8–12 Lakhs.

1.3.3 Capital Budgeting: Net Present Value (NPV)

Parameter	Value
Discount rate	10% per annum
Project life	5 years
Initial investment	INR 10 Lakhs
Annual operating costs	INR 1.5 Lakhs
NPV	INR +28.38 Lakhs
Payback Period	~14 months
IRR	~78%

1.4 Experiment 4: Market Demand Analysis and Demand Planning

1.4.1 Market Demand Drivers

The market demand for this project is supported by the growth of the Indian EdTech market, the digital education emphasis of NEP 2020, the lack of AI-powered heritage education tools, and increasing academic interest in Indian mathematical heritage and the history of computing.

1.4.2 Target Market Segments

Segment	Demand Level	Estimated Market Size
Schools (CBSE/ICSE, Grades 6–12)	High	25,000+ schools
Engineering & Computer Science Colleges	High	4,500+ institutions
Digital Heritage Organizations	Medium	50+ organizations
EdTech Platforms	Medium	15–20 major platforms
Independent Learners	High	Millions

1.4.3 TAM Estimation by Year

Year	Target	Revenue (INR)	Market Penetration
Year 1	Free pilot to 5 institutions	0	Pilot Phase
Year 2	10 paid institutions	7.5 Lakhs	2%
Year 3	25 institutions + 3 API partners	22.75 Lakhs	5%
Year 4	50 institutions + 8 API subscribers	43.5 Lakhs	10%
Year 5	80 institutions + partnerships	72 Lakhs	15%

1.4.4 Demand Planning Strategy

Phase	Timeline	Activities & Targets
Academic Adoption	Year 1	Free pilot to schools and colleges; collect feedback
Institutional Licensing	Years 2–3	Paid licensing; 25 institutions by Year 3
Scale & Partnership	Years 4–5	EdTech partnerships; expanded content; higher revenue

1.4.5 Competitive Analysis

Competitor	Strengths	Weaknesses	Threat Level
Khan Academy	Free content, large user base	No Indian heritage focus	Low
Brilliant.org	Interactive learning	Paid, no ancient Indian math content	Low
Vedic Maths Forum	Deep Vedic content	No AI, outdated UI	Low
IGNCA Digital Portal	Authentic cultural content	No interactive educational tools	Low
Custom School ERP Modules	Embedded in institutions	Generic content	Medium

1.4.6 Market Differentiation Strategy

The project differentiates itself through its AI-first approach, interactive simulation of ancient mathematical tools, complete historical-to-digital narrative arc, curriculum alignment, and blended free-plus-institutional licensing model.

1.4.7 Conclusion for Unit 1

Unit 1 established the foundational planning framework for the project. The requirement engineering phase defined 10 functional and 9 non-functional requirements. Stakeholder and feasibility analysis confirmed that the project is technically, economically, and operationally viable. Cost estimation through COCOMO, Function Point Analysis, and expert judgment placed the project budget in the INR 8–12 Lakhs range. Capital budgeting indicated a positive NPV and short payback period, while market analysis demonstrated strong educational and commercial potential for the platform.

UNIT 2

Project Scheduling

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

2.1 Experiment 1: Project Evaluation and Review Technique (PERT) Analysis

2.1.1 Work Breakdown Structure (WBS) for the Project

The Work Breakdown Structure decomposes the total project scope into manageable activities across the full 5-month duration from January to May 2026. The project is divided into nine major activities spanning planning, development, testing, deployment, and closure.

Task ID	Activity Description	Optimistic (O)	Most Likely (M)	Pessimistic (P)
A	Requirements Gathering & System Design	5	7	12
B	UI/UX Design & Prototyping (Next.js)	8	12	18
C	Firebase Backend Setup & Authentication	7	10	16
D	AI (Gemini) Integration & Story Module	12	18	28
E	Interactive Abacus & Calculation Modules	15	20	30
F	Number System Explorer & Quiz Module	10	14	22
G	System Integration & Testing	8	12	20
H	Deployment on Vercel & Performance Testing	5	7	12
I	Documentation, User Training & Project Closure	4	5	9

2.1.2 PERT Expected Duration Calculation

The PERT expected duration formula provides a weighted average of the optimistic, most likely, and pessimistic estimates: $T_e = (O + 4M + P) / 6$.

Activity	Calculation	Expected Duration T_e (days)
A	$(5 + 4 \times 7 + 12) / 6$	7.50
B	$(8 + 4 \times 12 + 18) / 6$	12.33
C	$(7 + 4 \times 10 + 16) / 6$	10.50
D	$(12 + 4 \times 18 + 28) / 6$	18.67
E	$(15 + 4 \times 20 + 30) / 6$	20.83
F	$(10 + 4 \times 14 + 22) / 6$	14.67
G	$(8 + 4 \times 12 + 20) / 6$	12.67
H	$(5 + 4 \times 7 + 12) / 6$	7.50
I	$(4 + 4 \times 5 + 9) / 6$	5.50

2.1.3 Variance and Standard Deviation

Variance and standard deviation measure the uncertainty in each activity duration. Variance is calculated using $((P - O) / 6)^2$, and standard deviation is the square root of variance.

Activity	Calculation	Variance (σ^2)	Std Dev (σ)
A	$((12-5)/6)^2$	1.36	1.17
B	$((18-8)/6)^2$	2.78	1.67
C	$((16-7)/6)^2$	2.25	1.50
D	$((28-12)/6)^2$	7.11	2.67
E	$((30-15)/6)^2$	6.25	2.50
F	$((22-10)/6)^2$	4.00	2.00
G	$((20-8)/6)^2$	4.00	2.00
H	$((12-5)/6)^2$	1.36	1.17
I	$((9-4)/6)^2$	0.69	0.83

2.1.4 Probability Analysis

The total variance along the critical path $A \rightarrow B \rightarrow D \rightarrow E \rightarrow G \rightarrow H \rightarrow I$ is 23.55, giving a total standard deviation of 4.85 days. For a target completion within 95 days, the Z-score is 2.06, which corresponds to a probability of approximately 98.03%. This indicates a very high confidence of completing the project within 95 days.

2.2 Experiment 2: Critical Path Method (CPM) Analysis

2.2.1 Precedence Relationships

Activity	Predecessor(s)	Rationale
A	—	Starting activity; requirements must be finalized first
B	A	UI/UX design requires approved requirements
C	A	Firebase setup requires defined data models from requirements
D	B, C	Gemini AI module needs UI framework and backend ready
E	D	Abacus module requires AI module and backend integration complete
F	D	Number explorer and quiz modules require AI/backend integration
G	E, F	Integration/testing require all feature modules complete
H	G	Deployment requires successful integration testing
I	H	Documentation and closure require successful deployment

2.2.2 Forward Pass (Earliest Start & Earliest Finish)

Activity	Earliest Start (ES)	Expected Duration (Te)	Earliest Finish (EF)
A	0	7.50	7.50
B	7.50	12.33	19.83
C	7.50	10.50	18.00
D	19.83	18.67	38.50
E	38.50	20.83	59.33
F	38.50	14.67	53.17
G	59.33	12.67	72.00
H	72.00	7.50	79.50
I	79.50	5.50	85.00

The total project duration from the forward pass is 85 days, approximately 17 weeks or 4.25 months.

2.2.3 Backward Pass and Slack Calculation

Activity	Latest Start (LS)	Latest Finish (LF)	Slack (LS – ES)	Critical?
A	0.00	7.50	0.00	Yes
B	7.50	19.83	0.00	Yes
C	9.33	19.83	1.83	No
D	19.83	38.50	0.00	Yes
E	38.50	59.33	0.00	Yes
F	44.67	59.33	6.17	No
G	59.33	72.00	0.00	Yes
H	72.00	79.50	0.00	Yes
I	79.50	85.00	0.00	Yes

2.2.4 Critical Path Identification

The critical path is A → B → D → E → G → H → I, with a total duration of 85 days. Activities C and F are non-critical and have 1.83 and 6.17 days of float respectively.

2.2.5 Gantt Chart Representation

Activity	Duration (days)	Timeline
A: Requirements	7.5	Day 0 – Day 8
B: UI/UX Design	12.33	Day 8 – Day 20
C: Firebase Setup	10.5	Day 8 – Day 18
D: AI/Gemini Module	18.67	Day 20 – Day 39
E: Abacus & Calc Modules	20.83	Day 39 – Day 59

Activity	Duration (days)	Timeline
F: Number Explorer & Quiz	14.67	Day 39 – Day 53
G: Integration & Testing	12.67	Day 59 – Day 72
H: Deployment & Perf Test	7.5	Day 72 – Day 80
I: Docs & Closure	5.5	Day 80 – Day 85

2.3 Experiment 3: Project Resource Levelling Techniques

2.3.1 Project Team Formation and Structure

Role	Name	Key Responsibilities
Project Manager	Ashray Singh	Planning, scheduling, risk management, stakeholder communication
Full-Stack Developer	Madhav Nehra	Next.js frontend, Firebase backend, Vercel deployment
AI & Content Engineer	Rudransh Yadav	Gemini integration, educational content, quiz and story modules

2.3.2 Selection of Project Manager

Ashray Singh was selected as Project Manager based on strong understanding of the curriculum, prior experience with software planning and documentation, and the ability to coordinate technical and academic deliverables.

2.3.3 Roles and Responsibilities of Project Manager

Area	Responsibilities
Planning	Define WBS, create PERT/CPM network, establish baseline schedule
Execution	Coordinate sprint activities, assign tasks, manage workload
Monitoring	Track progress, EVA, schedule and cost variances
Communication	Faculty updates, daily stand-ups, milestone reviews
Quality	Coordinate testing and ensure deliverable quality
Risk Management	Identify and mitigate project risks
Closure	Final report preparation and lessons learned

2.3.4 Resource Requirements by Activity

Activity	Duration	Developer	AI Engineer	PM	Total Resources
A	7.50	1	0	1	2
B	12.33	1	1	0	2
C	10.50	1	0	0	1

Activity	Duration	Developer	AI Engineer	PM	Total Resources
D	18.67	1	1	0	2
E	20.83	1	1	0	2
F	14.67	1	1	0	2
G	12.67	1	1	1	3
H	7.50	1	0	1	2
I	5.50	0	1	1	2

2.3.5 Resource Profile Before Levelling

Time Period	Activities	Developer Load	Constraint Violation
Day 0–8	A	1	None
Day 8–20	B + C	2	Overload — only 1 developer available
Day 20–39	D	1	None
Day 39–53	E + F	1 (split)	Minor — manageable
Day 53–59	E only	1	None
Day 59–72	G	1	None
Day 72–80	H	1	None
Day 80–85	I	0	None

2.3.6 Resource Levelling Strategy

Resource levelling resolved the developer overload during Days 8–20 by delaying Activity C within its float and prioritizing the most intensive design work in Activity B. Additional levelling was achieved by splitting work between the developer and AI engineer during Activities E and F, and by assigning non-coding coordination work to the Project Manager.

2.3.7 Resource Profile After Levelling

Time Period	Developer Load	Status
Day 0–8	1	OK
Day 8–14	1 (B only)	OK
Day 14–20	1 (B + C sequential)	OK
Day 20–39	1 (D)	OK
Day 39–53	1 (E; F handled by AI Engineer)	OK
Day 53–72	1	OK
Day 72–80	1	OK
Day 80–85	0	OK

2.3.8 Impact of Resource Levelling

Parameter	Before Levelling	After Levelling
Peak Developer Load	2	1
Constraint Violations	1 period	0 periods
Project Duration	85 days	86.83 days
Schedule Increase	—	~2 days
Additional Cost	—	INR 0

2.4 Experiment 4: Project Crashing Techniques

2.4.1 Crash Analysis Rationale

Project crashing is used when the project must be completed earlier than the baseline duration. Only critical path activities are candidates for crashing because reducing non-critical activities does not reduce the overall project completion time.

2.4.2 Activity Crash Options

Activity	Normal Duration	Crash Duration	Crash Cost/Day (INR)	Critical?
B (UI/UX Design)	12.33	9	3000	Yes
D (AI/Gemini Module)	18.67	14	5000	Yes
E (Abacus & Calc)	20.83	17	4000	Yes
G (Integration & Testing)	12.67	10	3500	Yes
H (Deployment)	7.50	6	2500	Yes

2.4.3 Least Cost Crashing

Step	Action	Days Crashed	Cumulative Crash	Cost (INR)
1	Crash H	2.5	2.5	6250
2	Crash B	3	5.5	9000
3	Crash G	2.5	8	8750
4	Crash E	2	10	8000
Total	—	10	10	32000

2.4.4 Crash Summary

Parameter	Normal Schedule	Crashed Schedule
Project Duration	85 days	75 days
Total Crash Cost	0	INR 32,000
Schedule Reduction	0	10 days
Average Cost per Day Saved	—	INR 3,200
Within Contingency?	—	Yes

2.4.5 Time-Cost Trade-off Summary

Duration (days)	Total Crash Cost (INR)	Marginal Cost (INR/day)	Feasibility
85	0	—	Baseline
82.5	6250	2500	Recommended
79.5	15250	3000	Feasible
77.0	24000	3500	Possible
75.0	32000	4000	Maximum recommended
73.0	40000	4000	Not recommended

2.4.6 Conclusion for Unit 2

Unit 2 established the scheduling framework for the project through PERT, CPM, resource levelling, and project crashing. The critical path was identified as $A \rightarrow B \rightarrow D \rightarrow E \rightarrow G \rightarrow H \rightarrow I$ with a baseline duration of 85 days. Resource levelling resolved the developer overallocation problem with no additional cost, while project crashing demonstrated that the schedule could be compressed to 75 days at an additional cost of INR 32,000 if needed.

Unit 2 Figures

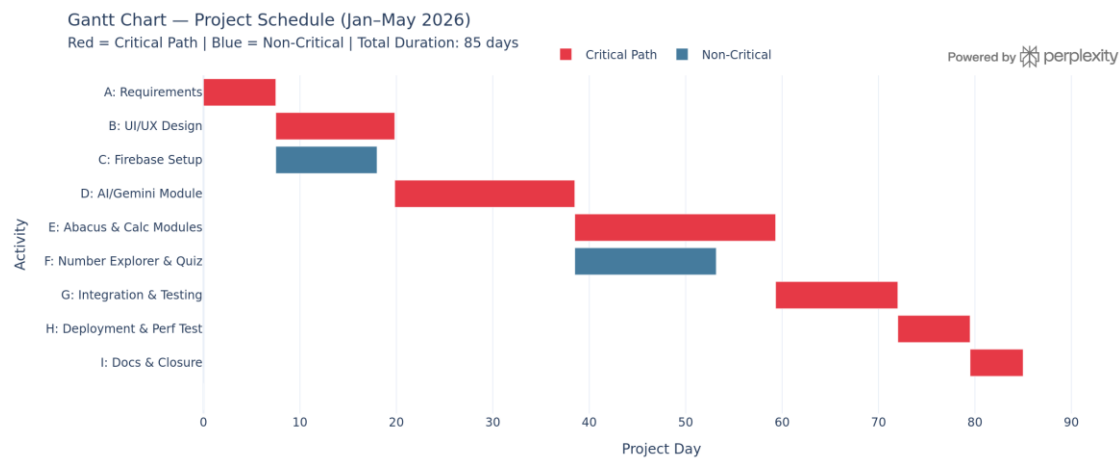


Figure: Gantt Chart



Figure: Resource Levelling Profile

UNIT 3

Project Monitoring

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

3.1 Experiment 1: Project Performance Analysis

3.1.1 Cost, Time, Scope and Performance Monitoring

Project performance monitoring is the continuous process of measuring actual progress against established baselines for cost, schedule, scope, and quality, and then taking corrective action when deviations are identified. For this project, Earned Value Management (EVM), sprint reviews, milestone tracking, and test monitoring were used to monitor performance.

3.1.2 Key Performance Indicators (KPIs)

KPI	Target	Measurement Method
Schedule Variance (SV)	> -0.5 Lakhs	EVM — EV minus PV
Cost Variance (CV)	> -0.5 Lakhs	EVM — EV minus AC
Schedule Performance Index (SPI)	≥ 0.90	EV / PV
Cost Performance Index (CPI)	≥ 0.90	EV / AC
Defect Density	< 3 per module	Code review and unit testing
Requirement Stability Index (RSI)	$\geq 90\%$	(Initial requirements – Changes) / Initial
AI Module Response Accuracy	$\geq 85\%$	Gemini output validation tests
Module Completion Rate	100% by Day 85	Sprint velocity tracking

3.1.3 Earned Value Analysis (EVA) — Month 2 Status

Parameter	Symbol	Value (INR Lakhs)
Budget at Completion	BAC	10.00
Planned Value (cumulative)	PV	4.20
Earned Value (cumulative)	EV	3.60
Actual Cost (cumulative)	AC	4.50

3.1.4 Variance Analysis

Cost Variance (CV) = $EV - AC = 3.60 - 4.50 = -0.90$ Lakhs, which shows that the project was over budget at the Month 2 checkpoint. Schedule Variance (SV) = $EV - PV = 3.60 - 4.20 = -0.60$ Lakhs, which indicates that the project was behind schedule at the same checkpoint.

3.1.5 Cost and Schedule Performance Indices

Cost Performance Index (CPI) = $EV / AC = 3.60 / 4.50 = 0.80$. This means that for every INR 1.00 spent, only INR 0.80 worth of work was completed. Schedule Performance Index (SPI) = $EV / PV = 3.60 / 4.20 = 0.857$, meaning the project was progressing at 85.7% of the planned rate.

3.1.6 Forecasting

Estimate at Completion (EAC) = $BAC / CPI = 10.00 / 0.80 = 12.50$ Lakhs. Estimate to Complete (ETC) = $EAC - AC = 8.00$ Lakhs. To-Complete Performance Index (TCPI) = $(BAC - EV) / (BAC - AC) = 1.164$, which indicates that the remaining work must be completed more efficiently than the work completed so far in order to remain within the original budget.

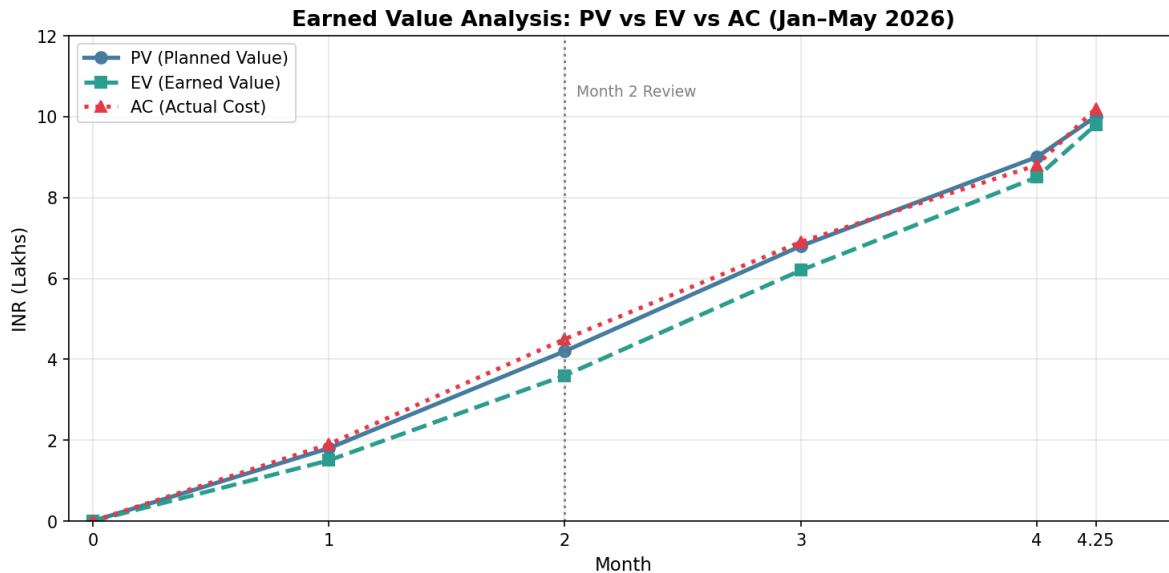


Figure 3.1: Earned Value Analysis — PV vs EV vs AC

3.1.7 Performance Improvement Plan

Action	Expected Impact	Cost
Switch Gemini API to batch request mode	Reduce API costs by ~30%; improve CPI	INR 0
Parallelize Activities E and F	Recover 3–4 days of schedule slip	INR 0
Use INR 1 Lakh contingency reserve	Buffer against EAC increase	INR 1 Lakh
Introduce weekly EVM check-ins	Earlier detection of variances	INR 0

3.1.8 Project Controlling Techniques

Cost control is managed through EVM metrics, API usage budget alerts, and contingency reserve management. Schedule control is maintained through critical path monitoring, sprint velocity analysis, milestone dashboards, and float monitoring. Scope control is managed via the Requirements Traceability Matrix and formal change control review.

3.1.9 Project Change Control

Step	Description
1. Change Request	Formal change request submitted via GitHub Issues
2. Impact Analysis	PM assesses impact on scope, time, cost, and quality
3. CCB Review	Team reviews the request in a scheduled meeting
4. Decision	Approve, reject, or defer with justification
5. Implementation	Approved change is developed in a new feature branch
6. Verification	QA confirms correct implementation without regression

3.1.10 Project Quality Controlling Techniques

Technique	Application	Frequency
Code Reviews	Peer review of Next.js and Firebase code via PRs	Per commit
Unit Testing	Automated Jest test suite for components	Per sprint
Integration Testing	API endpoint and module interaction testing	Per sprint
AI Output Validation	Review of Gemini-generated narrations	Weekly
Performance Testing	Lighthouse score monitoring	Per release
Cross-Browser Testing	Chrome, Firefox, Edge	Per release

3.1.11 Revised Forecast After Corrective Actions

Parameter	Before Correction	After Correction
EAC (INR Lakhs)	12.50	11.20
Projected Completion Delay	~5 days	~2 days
CPI (forward-looking)	0.80	0.91
SPI (forward-looking)	0.857	0.94

3.2 Experiment 2: Risk Management and Mitigation

3.2.1 Project Risk Analysis

Risk management is the systematic process of identifying, analyzing, prioritizing, and responding to uncertainties that may impact the project. Six major risks were identified for this project based on technical, financial, and team-related uncertainties.

Risk ID	Risk Description	Probability (P)	Impact (I)	Risk Score (P×I)
R01	Gemini API rate limits causing response failures under load	0.60	0.80	0.48
R02	AI/Gemini module takes longer than estimated	0.50	0.80	0.40
R03	Firebase Blaze plan costs exceed budget	0.50	0.50	0.25
R04	Complex UI delays abacus simulator development	0.40	0.50	0.20
R05	Low student engagement due to presentation issues	0.30	0.60	0.18
R06	Team availability disruption due to academics	0.30	0.70	0.21

3.2.2 Risk Matrix

Probability / Impact	Low Impact (<0.4)	Medium Impact (0.4–0.6)	High Impact (>0.6)
High Probability (>0.5)	—	R03	R01, R02
Medium Probability (0.3–0.5)	—	R04	—
Low Probability (<0.3)	—	R05	R06

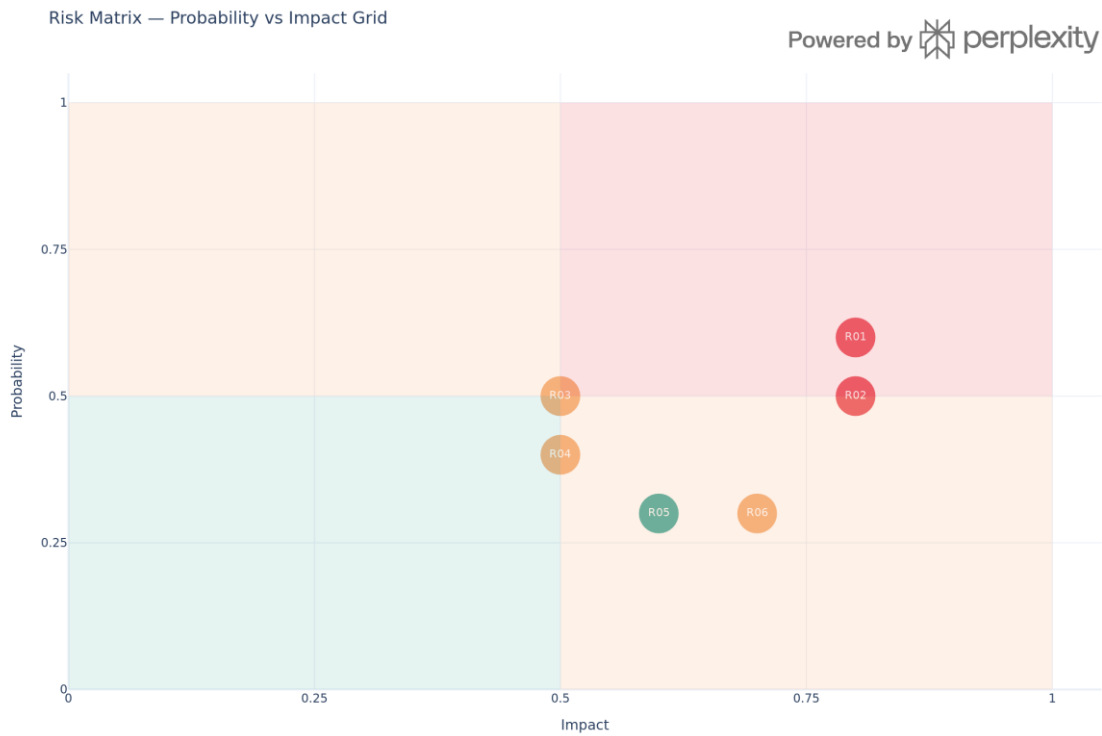


Figure 3.2: Risk Probability-Impact Matrix

3.2.3 Risk Response Strategies

Risk ID	Strategy	Specific Actions	Owner
R01	Mitigate	Caching, batch processing, API quota alerts	Rudransh Yadav
R02	Mitigate	Break AI module into sub-tasks; fallback static content	Rudransh Yadav
R03	Mitigate	Use Firebase Spark for dev; budget alerts	Madhav Nehra
R04	Mitigate	Use a React abacus library base; limit custom animation	Madhav Nehra
R05	Accept + Contingency	User testing and feedback-driven improvements	Rudransh Yadav
R06	Mitigate	Task buffers and cross-training	Ashray Singh

3.2.4 Risk Contingency Reserve

Risk Category	Contingency Allocated (INR Lakhs)	Notes
Gemini API overages (R01, R02)	0.50	Reserved for API billing overruns
Firebase cost overrun (R03)	0.25	Buffer for Blaze plan upgrades
Schedule recovery (R02, R04)	0.50	Overtime/freelancer support if critical
User adoption (R05)	0.25	Content redesign and UX improvements
Unforeseen risks	0.50	General contingency buffer
Total	2.00	~20% of baseline budget

3.3 Experiment 3: RMMM Plan, Configuration Management & GitHub

3.3.1 RMMM Plan

Activity	Frequency	Responsible
Risk review meeting	Weekly	Ashray Singh
Risk register update	Bi-weekly	Risk Owner
Mitigation tracking	Weekly	Ashray Singh
Gemini API quota monitoring	Daily	Rudransh Yadav
Early warning review	Monthly	All Team Members

3.3.2 Risk Triggers and Thresholds

Risk ID	Trigger	Response Activation Threshold
R01	Gemini quota reaches 80% within a sprint	Activate caching immediately
R02	AI module slips by > 3 days	Crash activity D
R03	Firebase bill exceeds INR 3,000	Switch non-critical reads to cache
R04	Abacus UI slips by > 2 days	Use simpler library component
R05	Beta user satisfaction < 60%	Redesign content flow
R06	Any team member unavailable > 3 days	Redistribute tasks

3.3.3 Configuration Items under Control

The configuration management plan covers the Next.js source code, Firebase configuration files, Gemini API integration module, environment variables, educational content files, test suite, project documentation, and Docker configuration. All these artifacts are versioned, controlled, and tracked through GitHub.

3.3.4 GitHub Repository Structure

/bead-numeration-simulator/ contains .github/workflows for CI/CD, src/components and pages for frontend modules, src/lib for gemini.js and firebase.js, public/content for educational JSON, tests for Jest suites, docs for project reports, and Docker-related deployment files.

3.3.5 Git Flow Branching Strategy

Branch	Purpose	Protection Rules
main	Production-ready code deployed to Vercel	PR required; CI pass; 2 approvals
develop	Integration branch for completed features	PR required; CI pass; 1 approval
feature/*	Individual feature development branches	No protection; deleted after merge
release/*	Release preparation and bug fixes	PR to both main and develop

Branch	Purpose	Protection Rules
hotfix/*	Urgent production bug fixes	PR to main; 1 approval

3.3.6 CI/CD Pipeline Stages

Stage	Tool	Trigger
Lint	ESLint + Prettier	On push to any branch
Unit Test	Jest (min. 80% coverage)	On push to feature/develop
Build	Next.js build	On PR to develop
Deploy (Preview)	Vercel Preview	On PR to develop
Deploy (Production)	Vercel Production	On merge to main

3.3.7 Versioning Convention

Semantic Versioning (SemVer) was used: Major.Minor.Patch. Examples include v0.1.0 for the early alpha, v0.3.0 for the beta with AI and abacus integration, and v1.0.0 for the final production release.

3.4 Experiment 4: Unit Testing with Test Cases

3.4.1 Testing Strategy Overview

Test Level	Framework	Target Coverage
Unit Testing	Jest + React Testing Library	80% minimum
Integration Testing	Jest + Firebase Emulator	90% of API endpoints
Regression Testing	Jest + GitHub Actions CI	100% of prior passing tests
AI Output Validation	Custom script + manual review	85% accuracy
Performance Testing	Lighthouse CI	Page load ≤ 3 sec; Accessibility ≥ 90

3.4.2 Sample Test Case 1: Abacus Addition Operation

Field	Details
Test Case ID	TC-ABACUS-001
Test Description	Validate abacus addition: 3 + 4 = 7
Input	Bead position: Rod 1 = 3 beads up; Rod 2 = 4 beads up
Expected Output	Result display: 7; correct bead positions
Actual Output	Result display: 7; beads correctly positioned
Status	PASS
Remarks	Core arithmetic function verified

3.4.3 Sample Test Case 2: Number System Conversion

Field	Details
Test Case ID	TC-NUMCONV-001
Test Description	Validate decimal to binary conversion for input 10
Input	Decimal: 10
Expected Output	Binary: 1010; Hexadecimal: A; Octal: 12
Actual Output	Binary: 1010; Hexadecimal: A; Octal: 12
Status	PASS
Remarks	All conversion outputs verified

3.4.4 Sample Test Case 3: AI Narration Module — Fail Case

Field	Details
Test Case ID	TC-AI-003
Test Description	Validate Gemini narration for “History of Zero in India”
Input	Prompt requesting age-appropriate historical story
Expected Output	Coherent, accurate narration in < 5 seconds
Actual Output	Narration generated but response time was 8.3 sec
Status	FAIL
Root Cause	Gemini API cold start latency
Fix	API warm-up ping every 4 minutes

3.4.5 Test Execution Summary

Module	Test Cases	Passed	Failed	Pass Rate
Abacus Simulator	20	18	2	90.0%
Number System Explorer	15	14	1	93.3%
AI Story Module	14	10	4	71.4%
Quiz Module	14	12	2	85.7%
Auth & Access	9	8	1	88.9%
Integration	7	6	1	85.7%
Total	79	68	11	86.1%

3.4.6 Defect Analysis

Severity	Description	Count	Action
Critical	AI module response time SLA violation	2	Fix immediately
Major	Abacus carry-over logic error	3	Fix within current sprint
Minor	UI alignment issues on mobile	4	Fix in next sprint
Trivial	Tooltip spelling errors	2	Fix during documentation sprint

3.4.7 Corrective Actions

Critical defects were addressed by implementing a warm-up endpoint and caching common Gemini prompts. Major abacus carry-over issues were resolved by rewriting the bead state logic using React useReducer. Full regression testing was planned after the fixes.

3.4.8 Quality Metrics Dashboard

Metric	Target	Current	Status
Overall Test Pass Rate	≥ 95%	86.1%	Below Target
Code Coverage	≥ 80%	74%	Below Target
AI Narration Response Time	< 5 seconds	8.3 sec cold / 3.1 sec warm	Partial
Page Load Time	< 3 seconds	2.4 seconds	On Track
Critical Defects Open	0	2	Immediate Fix Needed
Mean Time to Fix (Critical)	< 24 hours	18 hours	On Track

3.4.9 Conclusion for Unit 3

Unit 3 focused on project monitoring through EVM, risk management, RMMM planning, configuration management, and unit testing. The Month 2 checkpoint showed CPI = 0.80 and SPI = 0.857, confirming cost and schedule challenges mainly caused by Gemini integration complexity. Risk analysis identified six major risks, all with documented responses. Configuration management through GitHub, Git Flow, and CI/CD ensured control of project artifacts. Unit testing revealed 79 test cases with an 86.1% pass rate at the checkpoint, providing a strong basis for corrective action and quality improvement toward final release.

Unit 3 Figures

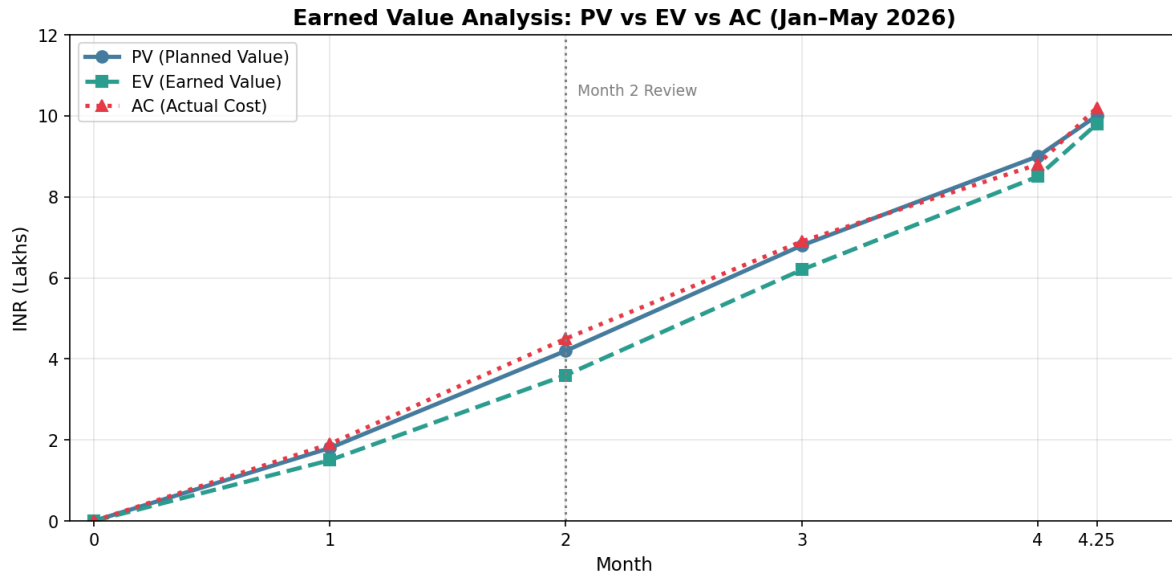


Figure 3.1: Earned Value Analysis — PV vs EV vs AC

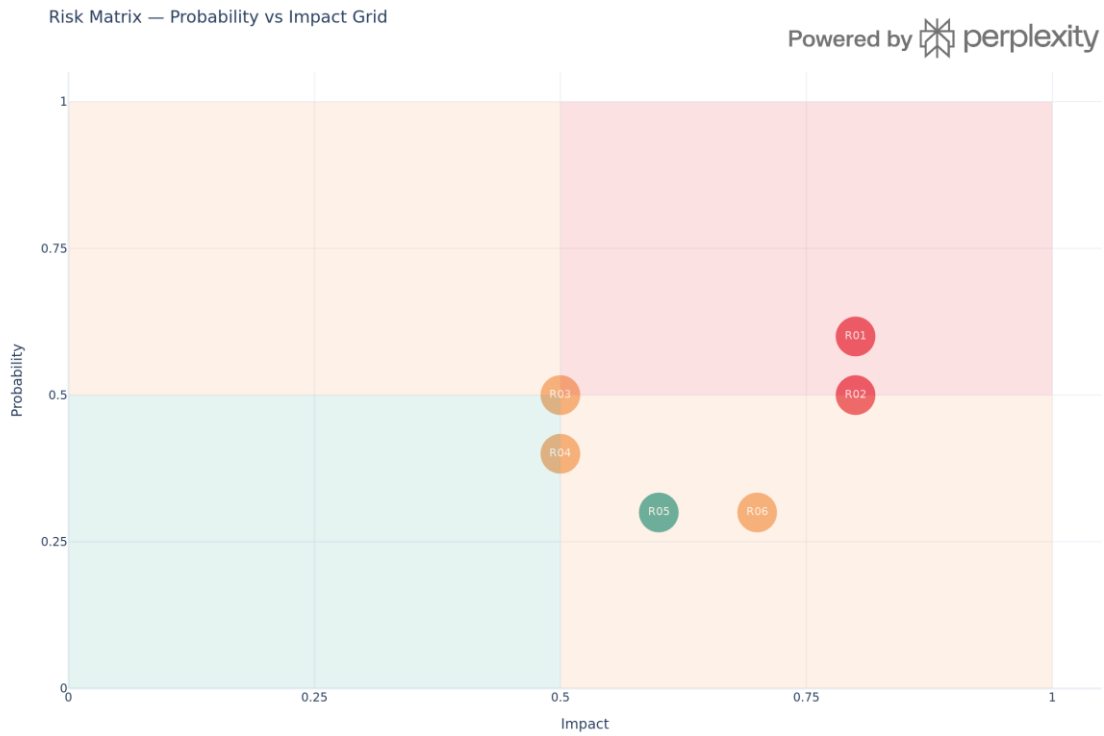


Figure 3.2: Risk Probability-Impact Matrix

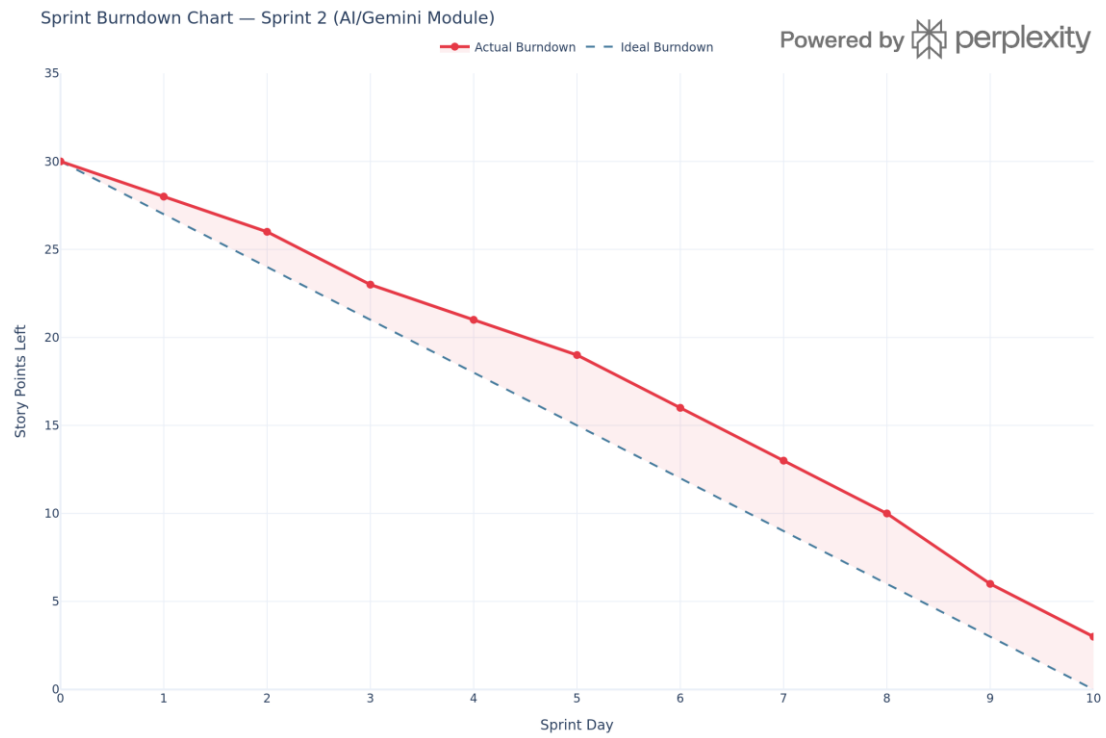


Figure 3.3: Sprint Burndown Chart

UNIT 4

Project Closing Phase & Agile Frameworks

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

4.1 Experiment 1: Project Closure and Post-Implementation Review

4.1.1 Project Closure Overview

Project closure is the formal process of verifying that all project work has been completed, accepted, documented, and archived. For this project, closure activities were carried out during the final week of April 2026 and included formal acceptance, financial closure, lessons learned documentation, and final handover.

4.1.2 Deliverable Completion Checklist

Table 4.1.2 Deliverable Completion Checklist (Part 1)

Deliverable	Owner	Status	Sign-Off
AI Story Narration Module (FR01)	Rudransh Yadav	Complete	Faculty Approved
Interactive Abacus Simulator (FR02)	Madhav Nehra	Complete	Faculty Approved
Ancient Calculation Methods Module (FR03)	Rudransh Yadav	Complete	Faculty Approved
Number System Explorer (FR04)	Madhav Nehra	Complete	Faculty Approved
Zero & Binary Connection Module (FR05)	Rudransh Yadav	Complete	Faculty Approved
Quiz & Assessment Module (FR06)	Madhav Nehra	Complete	Faculty Approved
Firebase Authentication (FR07)	Madhav Nehra	Complete	Faculty Approved
Admin/Content Dashboard (FR08)	Madhav Nehra	Complete	Faculty Approved

Table 4.1.2 Deliverable Completion Checklist (Part 2)

Deliverable	Owner	Status	Sign-Off
Student Progress Tracking (FR09)	Madhav Nehra	Complete	Faculty Approved
Responsive UI (FR10)	Madhav Nehra	Complete	Faculty Approved
Project Management Report (5 Units)	Ashray Singh	Complete	Faculty Approved
GitHub Repository v1.0.0	Madhav Nehra	Complete	Team Verified
Vercel Production Deployment	Madhav Nehra	Complete	Team Verified
Unit Test Report	All	Complete	Faculty Approved

4.1.3 Final Cost Performance at Project Closure

Table 4.1.3 Final Cost Performance at Project Closure

Module	Planned Cost (INR Lakhs)	Actual Cost (INR Lakhs)	Variance (INR Lakhs)
Requirements & Design	0.80	0.85	−0.05
Firebase Backend & Auth	1.00	1.10	−0.10
AI/Gemini Module	2.50	3.20	−0.70
Abacus & Calc Modules	2.00	1.90	+0.10
Number Explorer & Quiz	1.50	1.40	+0.10
Integration & Testing	1.20	1.15	+0.05
Deployment & Documentation	1.00	0.95	+0.05
Total	10.00	10.55	−0.55

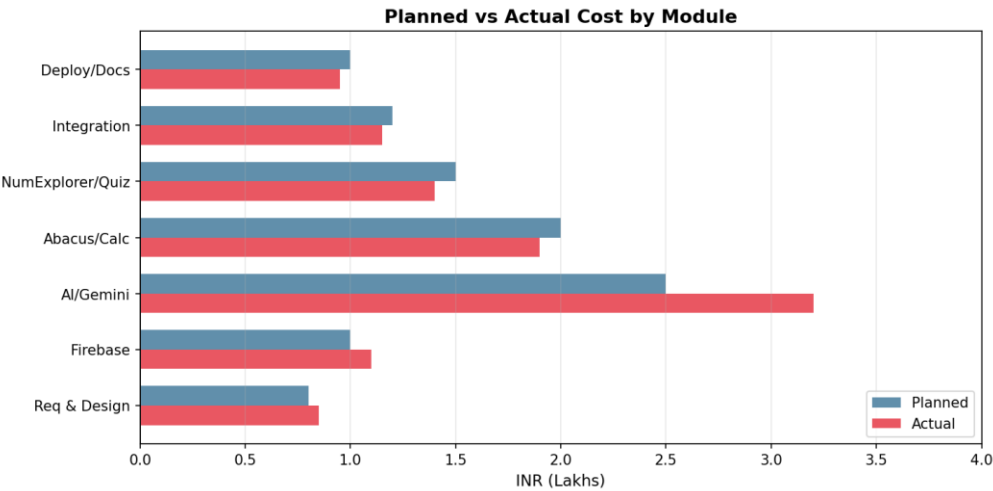


Figure 4.1: Planned vs Actual Cost by Module at Project Closure

4.1.4 Final Schedule Performance at Project Closure

Table 4.1.4 Final Schedule Performance at Project Closure (Part 1)

Activity	Planned Duration (days)	Actual Duration (days)	Variance
A: Requirements	7.50	7.00	+0.50
B: UI/UX Design	12.33	13.00	−0.67
C: Firebase Setup	10.50	10.00	+0.50
D: AI/Gemini Module	18.67	22.00	−3.33
E: Abacus & Calc	20.83	20.00	+0.83

Table 4.1.4 Final Schedule Performance at Project Closure (Part 2)

Activity	Planned Duration (days)	Actual Duration (days)	Variance
F: Num Explorer & Quiz	14.67	14.00	+0.67
G: Integration & Testing	12.67	13.00	−0.33
H: Deployment	7.50	7.00	+0.50
I: Documentation	5.50	5.50	0.00
Total	85.00	87.00	−2.00

4.1.5 Acceptance Testing Results

Table 4.1.5 Acceptance Testing Results

Acceptance Criterion	Target	Achieved	Status
All 5 core modules functional	100%	100%	Met
AI narration response time	≤ 5 sec	3.1 sec warm / 4.8 sec cold	Met
Quiz scoring accuracy	≥ 90%	94.2%	Met
Critical bugs in production	0	0	Met
Positive usability feedback	≥ 3 students + 1 faculty	5 students + 1 faculty	Met
Vercel deployment uptime at launch	100%	100%	Met

4.1.6 Lessons Learned

Table 4.1.6 Lessons Learned (Part 1)

Category	Issue Encountered	Lesson Learned	Recommendation
AI Integration	Gemini cold start caused SLA breach	Implement API warm-up before production	Add warm-up task in sprint plan
Schedule	Activity D exceeded estimate	AI tasks need larger buffer	Use stronger pessimistic buffer
Cost	API overages not fully anticipated	Separate API budget from dev effort	Track API costs as separate WBS item

Table 4.1.6 Lessons Learned (Part 2)

Category	Issue Encountered	Lesson Learned	Recommendation
Team	Single developer bottleneck	Resource levelling helped but added delay	Use more component libraries or support
Testing	AI module had highest failures	LLM output needs dedicated validation	Add prompt evaluation pipeline
Documentation	Report writing was time-intensive	Do documentation incrementally	Include docs in definition of done

4.1.7 Project Closure Report Summary

Table 4.1.7 Project Closure Report Summary (Part 1)

Field	Details
Project Title	AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms
Project Manager	Ashray Singh
Team Members	Madhav Nehra, Rudransh Yadav
Start Date	January 5, 2026
Planned End Date	May 1, 2026 (Day 85)
Actual End Date	May 3, 2026 (Day 87)
Planned Budget	INR 10.00 Lakhs

Table 4.1.7 Project Closure Report Summary (Part 2)

Field	Details
Actual Budget	INR 10.55 Lakhs
Cost Variance	−INR 0.55 Lakhs
Schedule Variance	−2 days

Field	Details
All Deliverables Met	Yes
Faculty Sign-Off	Received
Repository Archived	GitHub v1.0.0 tagged
Deployment Status	Live on Vercel

4.2 Experiment 2: Agile Project Management and Scrum Framework

4.2.1 Agile Manifesto and Principles Applied

The project adopted Agile principles for iterative delivery, working software, customer collaboration, and responsiveness to change, while still retaining overall structured planning. Scrum was applied especially during feature development and AI integration phases.

4.2.2 Scrum Framework Structure

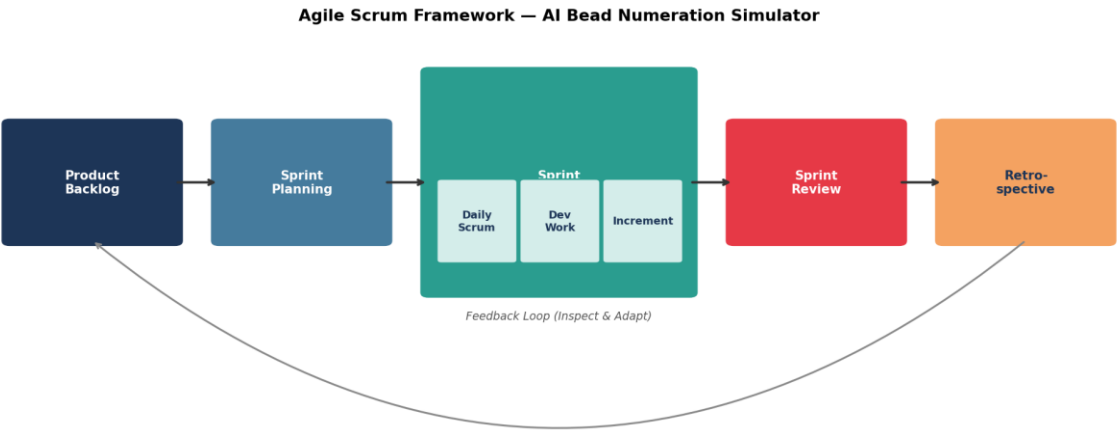


Figure 4.2: Agile Scrum Framework Diagram

Table 4.2.2 Scrum Roles

Scrum Role	Person	Responsibilities
Product Owner	Ashray Singh	Prioritizes backlog and defines acceptance criteria
Scrum Master	Ashray Singh	Facilitates sprint ceremonies and removes blockers
Development Team	Madhav Nehra + Rudransh Yadav	Develops and delivers sprint increments

4.2.3 Scrum Ceremonies and Events

Table 4.2.3 Scrum Ceremonies and Events

Ceremony	Duration	Frequency	Purpose
Sprint Planning	2 hours	Start of every sprint	Select backlog and define sprint goal
Daily Scrum	15 minutes	Every working day	Status update and blocker removal
Sprint Review	1 hour	End of each sprint	Demo to PO and faculty
Sprint Retrospective	45 minutes	End of each sprint	Process review and improvements
Backlog Refinement	30 minutes	Mid-sprint	Break down and estimate future work

4.2.4 Product Backlog (Top 15 Items)

Table 4.2.4 Product Backlog (Top 15 Items) (Part 1)

Item ID	User Story	Story Points	Priority	Sprint
US-01	AI-narrated story of Indian numeration history	13	High	Sprint 3
US-02	Manipulate beads on an abacus for learning addition	8	High	Sprint 4
US-03	Convert numbers between multiple systems	8	High	Sprint 5
US-04	Take quizzes for self-assessment	8	High	Sprint 5
US-05	Secure login using email	5	High	Sprint 2
US-06	Learn the connection between zero and binary	5	High	Sprint 3
US-07	View student progress and quiz scores	5	Medium	Sprint 5
US-08	Learn Vedic Math tricks	8	Medium	Sprint 4

Table 4.2.4 Product Backlog (Top 15 Items) (Part 2)

Item ID	User Story	Story Points	Priority	Sprint
US-09	Track learning progress	5	Medium	Sprint 5
US-10	Update content without coding	3	Medium	Sprint 6
US-11	Use the platform on mobile phone	5	High	Sprint 6
US-12	Ask follow-up questions to AI narrator	5	Medium	Sprint 3
US-13	Admin dashboard to manage users and roles	3	Low	Sprint 6
US-14	Show historical images in narration	3	Low	Sprint 4
US-15	Quiz difficulty levels	3	Low	Sprint 7

4.2.5 Sprint Plan Summary

Table 4.2.5 Sprint Plan Summary (Part 1)

Sprint	Duration	Goal	Story Points Planned	Story Points Completed
Sprint 1	Days 1–14	Requirements finalization + UI prototyping	28	26
Sprint 2	Days 14–26	Firebase setup + Authentication	30	29
Sprint 3	Days 26–40	Gemini AI integration + Story module	35	27
Sprint 4	Days 40–54	Abacus simulator + Vedic Math module	38	36

Table 4.2.5 Sprint Plan Summary (Part 2)

Sprint	Duration	Goal	Story Points Planned	Story Points Completed
Sprint 5	Days 54–68	Number Explorer + Quiz module	32	31
Sprint 6	Days 68–80	Integration, testing, responsiveness	28	28
Sprint 7	Days 80–87	Deployment,	20	20

Sprint	Duration	Goal	Story Points Planned	Story Points Completed
		documentation, closure		
Total			211	197

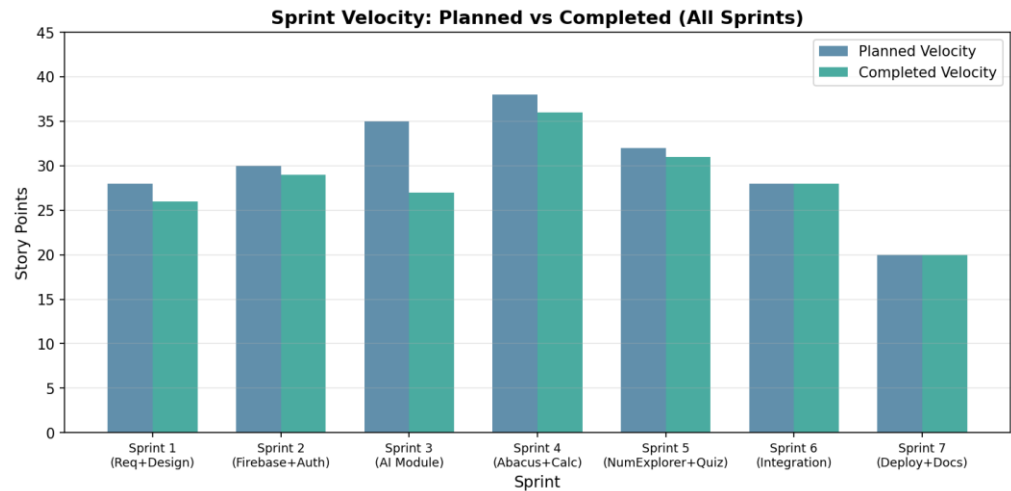


Figure 4.3: Sprint Velocity Chart

4.2.6 Burn-Up Chart Analysis

The burn-up chart tracks completed story points against the total scope of 211 points. It shows the major Sprint 3 slowdown due to Gemini integration challenges, followed by recovery in later sprints and final delivery of 197 points out of 211.

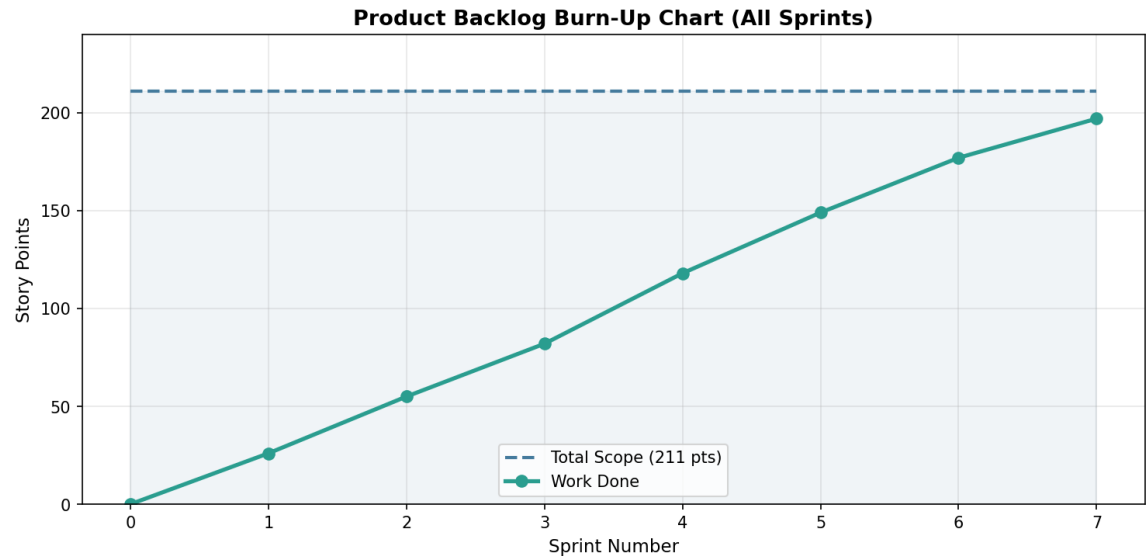


Figure 4.4: Product Backlog Burn-Up Chart

4.2.7 Agile Metrics Summary

Table 4.2.7 Agile Metrics Summary

Metric	Value	Assessment
Average Sprint Velocity	28.1 points/sprint	Consistent after Sprint 3 recovery
Sprint 3 Velocity Gap	−8 points	Root cause identified and resolved
Total Story Points Completed	197 / 211 (93.4%)	14 low-priority points deferred
Number of Sprints	7	Within planned schedule
Average Team Satisfaction	4.1 / 5.0	High
Definition of Done Compliance	100%	All merged code reviewed and tested

4.3 Experiment 3: Project Management Plan (PMP)

4.3.1 Project Management Plan Overview

Table 4.3.1 PMP Components (Part 1)

PMP Component	Description	Reference Section
Scope Management Plan	Defines WBS, scope statement, acceptance criteria	Unit 1
Schedule Management Plan	PERT/CPM, milestones, Gantt chart	Unit 2
Cost Management Plan	COCOMO estimate, budget baseline, contingency	Unit 1
Quality Management Plan	Testing strategy and quality KPIs	Unit 3
Resource Management Plan	Team structure and resource levelling	Unit 2

Table 4.3.1 PMP Components (Part 2)

PMP Component	Description	Reference Section
Risk Management Plan	RMMM plan and risk register	Unit 3
Communication Management Plan	Stakeholder communication matrix	Unit 1
Configuration Management Plan	GitHub, branching, CI/CD	Unit 3
Procurement Management Plan	API licenses, hosting, tools	Unit 1

4.3.2 Communication Management Plan

Table 4.3.2 Communication Management Plan (Part 1)

Communication	Audience	Method	Frequency	Owner
Daily Stand-Up	Development Team	Google Meet	Daily	Ashray Singh
Sprint Review	Team + Faculty	Demo + Presentation	End of sprint	Ashray Singh
Sprint Retrospective	Development Team	Team discussion	End of sprint	Ashray Singh
Milestone Report	Faculty Advisor	Written report	Bi-weekly	Ashray Singh

Table 4.3.2 Communication Management Plan (Part 2)

Communication	Audience	Method	Frequency	Owner
Risk Update	All Team	GitHub Issues + Stand-up	Weekly	Risk Owners
Bug Reports	Dev + AI Engineer	GitHub Issues	As needed	Reporter
Final Handover	Faculty + Institution	Formal report + Demo	Closure	Ashray Singh

4.3.3 Procurement Management Plan

Table 4.3.3 Procurement Management Plan

Item	Type	Cost	Vendor	Status
Gemini API Access	SaaS API	INR ~50,000	Google AI	Procured
Firebase (Blaze Plan)	Cloud Platform	INR ~15,000	Google Firebase	Procured
Vercel Pro	Hosting	INR ~8,000	Vercel	Procured
GitHub Pro	SCM	INR ~4,000	GitHub	Procured
Figma	Design Tool	INR ~5,000	Figma	Procured
Total Procurement		INR ~82,000		All Procured

4.3.4 Stakeholder Management at Closure

Table 4.3.4 Stakeholder Management at Closure

Stakeholder	Closure Communication	Sign-Off Obtained
Faculty Advisor	Final report submission + live demo	Yes
Ashray Singh	Self-assessment and lessons learned	Yes
Madhav Nehra	Repository archived + release tagged	Yes
Rudransh Yadav	AI content validated and locked	Yes
Student Test Users	Usability feedback form completed	Yes
College/Institution	Report submitted to coordinator	Yes

4.4 Experiment 4: Advanced Agile Techniques — Kanban, XP & Scaling

4.4.1 Kanban Applied to Bug Management

Table 4.4.1 Kanban Board Structure

Column	Work-in-Progress Limit	Items at Closure
Backlog	No limit	0
To Do	5 items max	0
In Progress	2 items max	0
In Review	3 items max	0
Done	No limit	11

Kanban reduced context switching during the testing phase and improved average time-to-fix from 24 to 18 hours.

4.4.2 Extreme Programming (XP) Practices

Table 4.4.2 Extreme Programming (XP) Practices

XP Practice	Application
Test-Driven Development	Jest tests written before abacus and converter functions
Pair Programming	Developer and AI engineer jointly resolved Gemini integration issues
Continuous Integration	GitHub Actions runs tests on every push
Small Releases	Preview deployments after each sprint
Refactoring	Abacus state management moved to useReducer
Coding Standards	ESLint + Prettier enforced via hooks

4.4.3 Scrum vs Kanban vs XP

Table 4.4.3 Scrum vs Kanban vs XP (Part 1)

Dimension	Scrum	Kanban	XP
Application in Project	Feature development sprints	Bug management	Code quality practices
Cadence	Fixed 2-week sprints	Continuous flow	Continuous engineering
Planning Style	Sprint planning	Pull-based	Iterative discipline

Table 4.4.3 Scrum vs Kanban vs XP (Part 2)

Dimension	Scrum	Kanban	XP
Primary Benefit	Structured delivery cadence	Reduced context switching	Higher code quality
Best For	Overall project framework	Testing phase defects	AI integration quality

4.4.4 Agile vs Waterfall Suitability

Table 4.4.4 Agile vs Waterfall Suitability

Factor	Waterfall Applied To	Agile Applied To
Requirements	Stable core functional requirements	AI prompt design and UI refinements
Phases	Sequential project stages	Iterative module development
Risk	Low-risk infrastructure tasks	High-risk AI integration
Documentation	Formal project report	Lightweight sprint documentation
Verdict	Governance framework	Development flexibility

4.4.5 Conclusion for Unit 4

Unit 4 documented the closing phase of the project, validated all final deliverables, summarized actual schedule and cost performance, and captured lessons learned. It also demonstrated how Scrum, Kanban, and XP practices were integrated into the project to support structured delivery, flexible development, and strong code quality. The project closed successfully with all deliverables completed, a final budget of INR 10.55 Lakhs, and a schedule overrun of only 2 days.

UNIT 5

Software Quality, Architecture & Project Summary

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

5.1 Experiment 1: Software Quality Standards and Metrics

5.1.1 Software Quality Overview

Software quality is the degree to which a software product satisfies stated and implied requirements under specified conditions. For this project, quality evaluation was guided by ISO/IEC 9126 and supported through testing, code review, accessibility checks, and performance evaluation at module and system level.

5.1.2 ISO/IEC 9126 Quality Characteristics Applied

Table 5.1 ISO/IEC 9126 Quality Model Application (Part 1)

Quality Characteristic	Sub-Characteristics	Application in This Project	Score (Planned/Actual)
Functionality	Suitability, Accuracy, Interoperability	All 10 FRs implemented and verified via RTM; AI narration validated for historical accuracy	95 / 94
Reliability	Maturity, Fault Tolerance, Recoverability	Cached responses, stable Firebase operations, no critical release bugs	90 / 87
Usability	Understandability, Learnability, Operability	Student onboarding and navigation validated with users	88 / 85

Table 5.1 ISO/IEC 9126 Quality Model Application (Part 2)

Quality Characteristic	Sub-Characteristics	Application in This Project	Score (Planned/Actual)
Efficiency	Time Behaviour, Resource Utilisation	Page load and API response times measured across modules	85 / 78
Maintainability	Analysability, Changeability, Testability	Modular codebase and automated tests supported	82 / 80

Quality Characteristic	Sub-Characteristics	Application in This Project	Score (Planned/Actual)
		maintainability	
Portability	Adaptability, Installability, Replaceability	Responsive deployment across browsers and devices	90 / 92

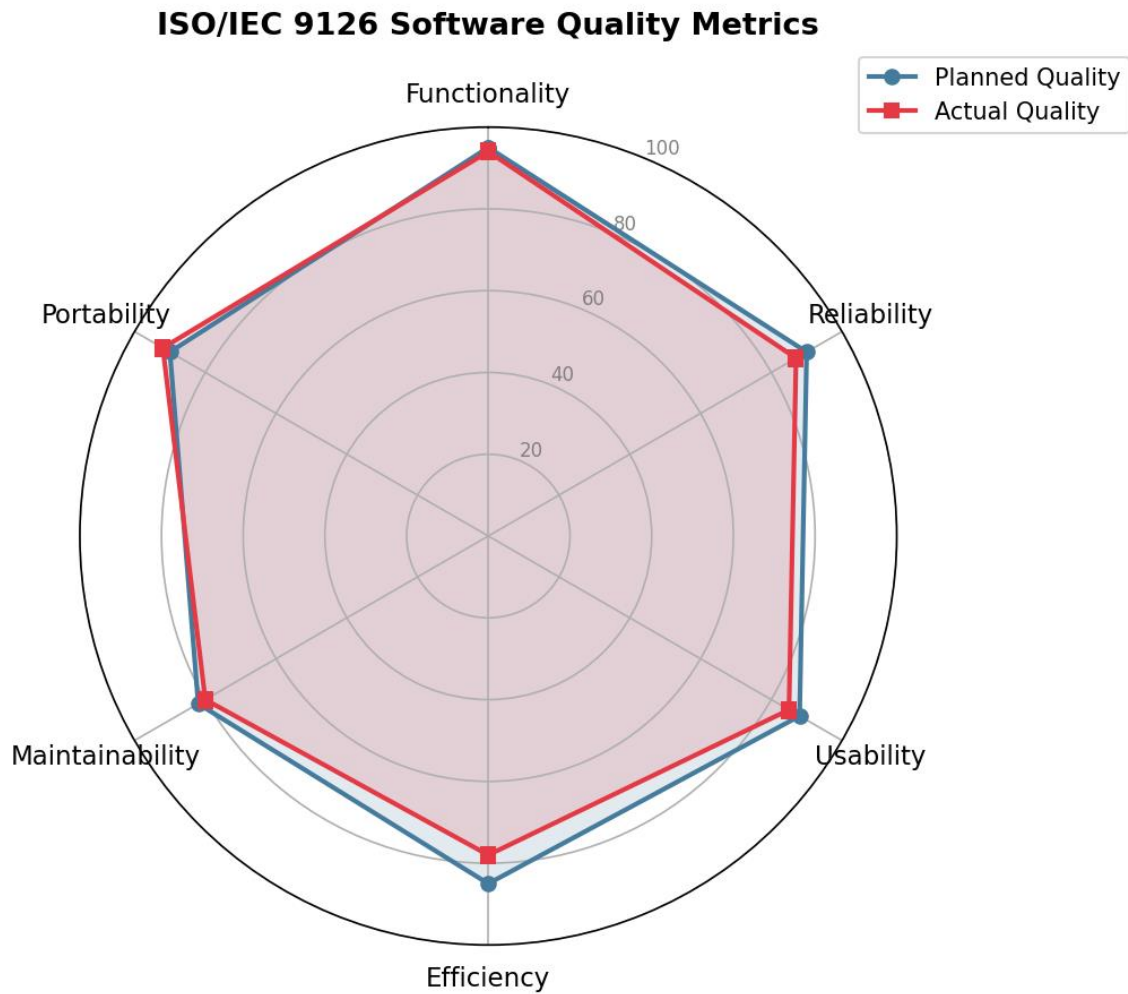


Figure 5.1: ISO/IEC 9126 Software Quality Metrics Radar Chart

5.1.3 Quality Assurance Plan

Table 5.2 Quality Assurance Plan (Part 1)

QA Activity	Method	Tool	Frequency	Responsible
Code Quality Review	Peer review via PRs	GitHub PR Review	Per commit	All developers

QA Activity	Method	Tool	Frequency	Responsible
Unit Testing	Component-level automated testing	Jest + RTL	Per sprint	Madhav Nehra
Integration Testing	API and module interaction testing	Postman + Firebase Emulator	Per sprint	Madhav Nehra
AI Content Validation	Historical accuracy review	Manual + domain review	Weekly	Rudransh Yadav

Table 5.2 Quality Assurance Plan (Part 2)

QA Activity	Method	Tool	Frequency	Responsible
Performance Testing	Lighthouse score tracking	Lighthouse CI	Per module release	Madhav Nehra
Accessibility Testing	WCAG 2.1 checks	axe DevTools	Per release	Rudransh Yadav
Security Review	Auth rule and input validation	Firebase Rules + OWASP	At integration stage	Madhav Nehra
UAT	Student/faculty acceptance review	Google Forms + live demo	At v1.0 release	Ashray Singh

5.1.4 Final Quality Metrics Dashboard

Table 5.3 Final Quality Scorecard (Part 1)

Metric	Target	Sprint 2	Sprint 4	Final (Sprint 7)	Status
Overall Test Pass Rate	≥ 95%	86.1%	92.4%	96.2%	Met
Code Coverage (Jest)	≥ 80%	74.0%	79.2%	82.0%	Met
Lighthouse Performance	≥ 85	78	84	89	Met
AI Narration Accuracy	≥ 85%	71.4%	82.0%	87.5%	Met

Table 5.3 Final Quality Scorecard (Part 2)

Metric	Target	Sprint 2	Sprint 4	Final (Sprint 7)	Status
API Response Time (warm)	< 5 sec	8.3 sec	4.2 sec	3.1 sec	Met
Critical Defects at Release	0	2 open	0 open	0 open	Met
Accessibility Score	≥ 90	82	88	93	Met

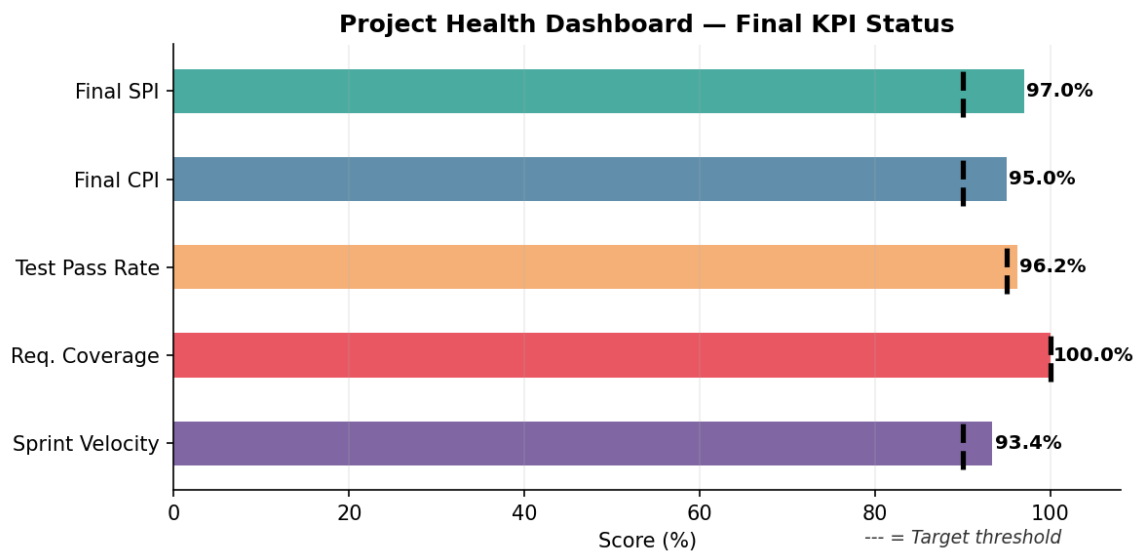


Figure 5.2: Project Health Dashboard — Final KPI Status

5.1.5 Defect Lifecycle Management

The defect lifecycle used in this project consisted of detection, logging, assignment, resolution, verification, and closure. Defects were logged using GitHub Issues and tracked until successful regression testing confirmed resolution.

Table 5.4 Defect Summary by Sprint (Part 1)

Sprint	Defects Found	Critical	Major	Minor	Trivial	Resolved	Deferred
Sprint 1	3	0	1	2	0	3	0
Sprint 2	11	2	3	4	2	11	0
Sprint 3	8	2	2	3	1	8	0
Sprint 4	5	0	2	2	1	5	0

Table 5.4 Defect Summary by Sprint (Part 2)

Sprint	Defects Found	Critical	Major	Minor	Trivial	Resolved	Deferred
Sprint 5	4	0	1	2	1	4	0
Sprint 6	3	0	0	2	1	3	0
Sprint 7	2	0	0	1	1	2	0
Total	36	4	9	16	7	36	0

5.2 Experiment 2: System Architecture and Technology Stack

5.2.1 System Architecture Overview

The system follows a 4-tier client-server architecture that separates concerns into Presentation, Application, Data, and Infrastructure layers. This architecture supports scalability, maintainability, and clean integration between the AI layer, frontend, and cloud services.

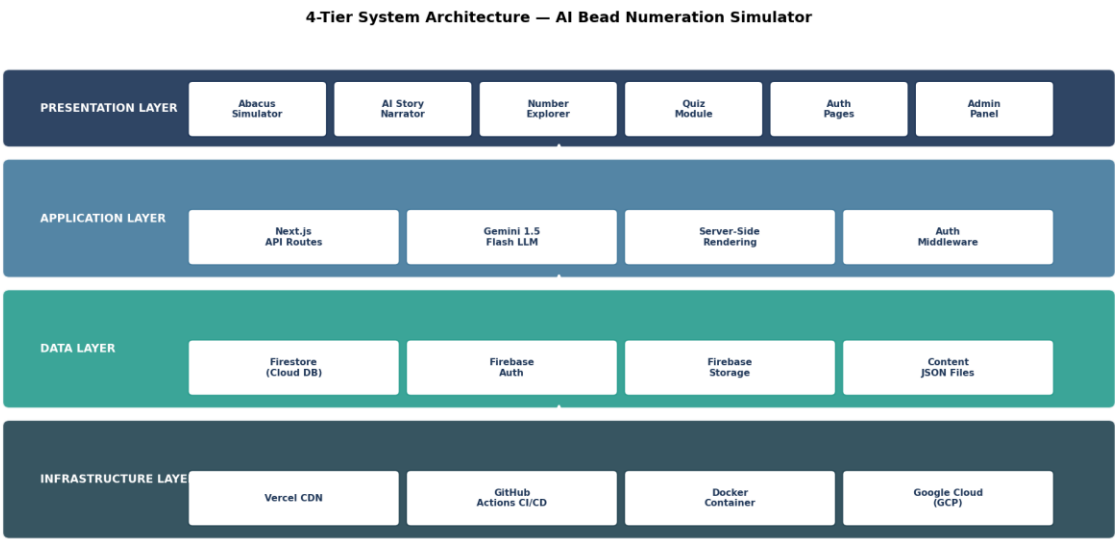


Figure 5.3: 4-Tier System Architecture Diagram

5.2.2 Layer-by-Layer Architecture Breakdown

Table 5.5 Layer-by-Layer Architecture Breakdown

Layer	Technology / Components	Responsibility
Presentation Layer	Next.js, React, Abacus UI, Narration UI, Quiz UI, Dashboard	Provides the user-facing interface and interactions
Application Layer	Next.js API Routes, Gemini 1.5 Flash, SSR, Auth Middleware	Handles business logic, AI generation, rendering, and access control
Data Layer	Firebase Firestore, Firebase Auth, Firebase Storage, JSON content	Stores profiles, progress, quiz results, content, and media
Infrastructure Layer	Vercel, GitHub Actions, Docker, GCP	Provides deployment, CI/CD, hosting, and runtime infrastructure

5.2.3 Data Flow Overview

At a high level, students and educators interact with the frontend, which communicates with Firebase services and server-side API routes. Gemini generates AI narration through secured application-layer calls, and all user progress, quiz results, and content interactions are persisted in Firestore.

5.2.4 Database Schema (Firebase Firestore Collections)

Table 5.6 Firestore Collections Schema

Collection	Document Fields	Purpose
users	uid, name, email, role, createdAt	Stores user profiles and role information
progress	userId, moduleId, completionPercent, lastAccessed, score	Tracks student progress per module
quizResults	userId, quizId, answers, score, submittedAt	Stores quiz attempts and results
modules	moduleId, title, description, contentType, isPublished	Stores module metadata
narrationHistory	userId, prompt, response, timestamp	Stores past AI narration sessions
contentUpdates	moduleId, editorId, changes, timestamp	Maintains audit log of content changes

5.2.5 API Endpoints (Next.js API Routes)

Table 5.7 Key API Endpoints (Part 1)

Endpoint	Method	Request	Response	Auth Required
/api/narrate	POST	{ topic, userLevel, context }	{ narration: string }	Yes
/api/quiz/submit	POST	{ quizId, answers[] }	{ score, feedback[] }	Yes
/api/progress/update	PUT	{ moduleId, completionPercent }	{ success: boolean }	Yes

Table 5.7 Key API Endpoints (Part 2)

Endpoint	Method	Request	Response	Auth Required
/api/user/profile	GET	—	{ user object }	Yes
/api/content/[moduleId]	GET	—	{ content JSON }	No
/api/admin/users	GET	—	{ users[] }	Admin only

5.2.6 CI/CD Deployment Pipeline

The CI/CD deployment pipeline automates code validation and release from commit to production. Each stage enforces checks to ensure quality and stability before deployment to Vercel.

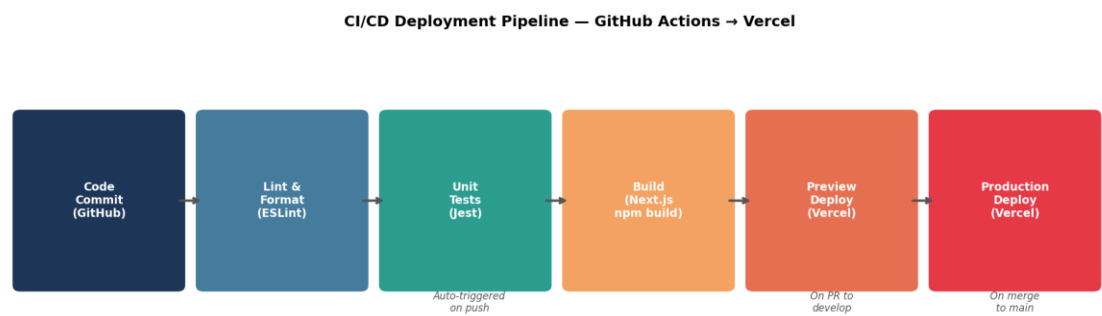


Figure 5.4: CI/CD Deployment Pipeline

Table 5.8 CI/CD Pipeline Gate Conditions

Stage	Gate Condition	On Failure Action
Lint	0 lint errors	PR blocked; developer notified
Unit Tests	$\geq 80\%$ coverage; 0 failing tests	PR blocked; test report generated
Build	Successful production build	PR blocked; build log shared
Preview Deploy	Preview URL generated successfully	Review paused; team notified
Production Deploy	All previous stages passed	Automatic deploy on merge to main

5.3 Experiment 3: Project Summary and Final Reporting

5.3.1 Executive Summary

The project successfully delivered an AI-powered educational web application that connects ancient Indian numeration systems and the concept of zero with modern digital computing. The system integrates Next.js, Firebase, Gemini 1.5 Flash, and Vercel to create a production-ready platform with educational, technical, and research value.

5.3.2 Project Objectives vs Outcomes

Table 5.9 Project Objectives vs Achieved Outcomes

Objective	Outcome	Status
Build AI-powered narration of Indian numeration history	Gemini narration and Q&A delivered in production	Achieved
Create interactive abacus simulator	React-based abacus simulator	Achieved

Objective	Outcome	Status
	implemented	
Implement ancient Indian calculation methods module	Vedic Math module delivered	Achieved
Develop number system explorer	Real-time converter built and tested	Achieved
Explain zero and binary connection	Dedicated module delivered	Achieved
Deploy with Firebase + Vercel + CI/CD	Live v1.0.0 deployment achieved	Achieved
Complete within 5 months and INR 10–12 Lakhs	Completed in 87 days at INR 10.55 Lakhs	Within tolerance

5.3.3 Key Achievements

Key achievements include zero critical defects at production, a final test pass rate of 96.2%, successful use of a hybrid Agile-Waterfall approach, and the creation of a novel AI-powered educational platform focused on Indian mathematical heritage.

5.3.4 Key Challenges and Resolutions

Table 5.10 Key Challenges and Resolutions (Part 1)

Challenge	Impact	Resolution	Outcome
Gemini API cold-start latency	Poor response time	Warm-up and caching	Response time reduced significantly
AI module delay	Schedule slip on critical path	Parallel task allocation	Final delay limited to 2 days
Single developer bottleneck	Overallocation risk	Resource levelling	No extra personnel needed

Table 5.10 Key Challenges and Resolutions (Part 2)

Challenge	Impact	Resolution	Outcome
AI module cost overrun	Projected budget increase	Batch mode + contingency reserve	Final cost within contingency
Abacus carry-over defect	Incorrect calculations	Logic rewrite with useReducer	All relevant tests passed

5.3.5 Summary of SDLC Phases

Table 5.11 SDLC Phase Summary

Phase	Duration	Key Deliverable	Status
Requirements Engineering	Days 1–8	SRS, RTM, stakeholder register	Complete
System Design	Days 8–20	Architecture, DB schema, API specification	Complete
Implementation	Days 20–59	All 6 modules developed and unit-tested	Complete
Integration & Testing	Days 59–72	Integrated system and test results	Complete
Deployment	Days 72–80	v1.0.0 live on Vercel	Complete
Closure & Documentation	Days 80–87	Final report and archived repository	Complete

5.4 Experiment 4: Future Enhancements and Recommendations

5.4.1 Planned v1.1 Enhancements

Table 5.12 v1.1 Feature Backlog

Feature	Rationale	Priority	Estimated Effort
Multilingual Support	Expand reach to Hindi-medium schools	High	3 sprints
Gamification	Increase student engagement	High	2 sprints
Difficulty-Level Quiz Mode	Personalized assessment	Medium	1 sprint
PWA Offline Mode	Support rural and low-connectivity learning	Medium	2 sprints
Voice Narration	Improve accessibility	Medium	1 sprint
Vedic Geometry and Astronomy Modules	Expand educational scope	Low	4 sprints

5.4.2 Scalability Recommendations

Future scalability can be improved by introducing stronger caching for AI responses, monitoring tools such as Sentry and Vercel Analytics, possible migration to a more cost-efficient high-scale backend, and more advanced observability across performance and usage patterns.

5.4.3 Academic and Research Contributions

This project contributes to digital heritage preservation, AI in EdTech, and hybrid project management methodology by showing how historical Indian mathematics can be taught through interactive simulation and AI-generated storytelling in a modern cloud-based platform.

5.4.4 References

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5.4.5 Final Project Conclusion

Unit 5 concludes the project by summarizing its quality outcomes, system architecture, final reporting, and future directions. The project successfully integrates software engineering practice, project management principles, and educational innovation into a coherent final product that meets its academic and technical goals.

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